Mapping Linked Genes

Biology 2e chapter 13 reading connection | Teacher guide / classroom preparation

Learning target

Do two traits assort independently, or is there evidence of linkage?

Timing and materials

Virtual first; one scoring class with prepared offspring or several weeks for live crosses.

Optional documented two-point Drosophila cross kit, culture vials and medium, dissecting microscopes, supplier-approved immobilization equipment, sorting brushes and scoring sheets. A photograph/data investigation can replace live handling.

Classroom route

1. Use an educational cross with known linked traits and parent genotypes. Follow its culture and handling instructions. Cross timing and selection of virgin females require advance preparation; allow several weeks.

2. Use heterozygous females for a recombination investigation: male Drosophila do not normally undergo meiotic crossing over. Record the parental combinations from the cross history.

3. Score offspring for both traits without selecting only unusual flies. Keep vial identity, ambiguous scores and total sample sizes. Estimate recombination percent; two traits estimate distance, not three-gene order.

Pilot the physical procedure with your own equipment. Follow school handling and disposal rules and the purchased kit instructions. The virtual numerical settings are not an equipment prescription.

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Hybrid or substitute assignment

Assign all virtual, all physical, or specified samples in each mode. The classroom entry button is available during collection. Students complete the same prediction and evidence questions. Record classroom time series on the worksheet; the online classroom form stores endpoint measurements and notes. Keep sources and measurement scales separate.

Guided virtual sequence

Students prepare each sample, inspect the apparatus or classify specimens, then explicitly collect data. All assigned trials are required before exploration. Reload pauses the clock and restores saved work. Downloads retain source labels; no automatic grade is submitted.

Expected reasoning

Gray/long and dark/short are parental; gray/short and dark/long are recombinant for this stated cross. Recombination percent = recombinant offspring / total x 100. Male Drosophila are not suitable heterozygous parents for this crossover assay. Two-point data do not determine three-gene order.

Feedback and scoring

Criterion	Points	Evidence
Prediction and controls	0-2	Testable claim; comparable samples and controlled variables.
Observations	0-2	Complete, labeled records with units and source.
Interpretation	0-2	Correct calculations or categorical comparisons; claim tied to evidence.
Limitations	0-2	Specific uncertainty and a useful follow-up investigation.

Score reasoning and data quality, not agreement of physical results with the simulation. Classification feedback is a practice aid and does not overwrite student scores.

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Model limitations

Original stylized flies and simulated offspring under an illustrative 18% recombination probability. No actual supplier stock dataset or viability effects are represented. Two-point mapping cannot establish three-gene order.

Troubleshooting

Confirm the preparation steps are finished before starting or collecting. Counting activities require every specimen plus a review; cell measurements require a focused 40x view; gels require staining and three size estimates; fluorescence labs require the viewer. Use the visible Collect data and continue button to advance.

For physical work, preserve unexpected observations, check the control and instrument setup, then repeat with fresh material where appropriate. Never replace measured results with the virtual answer.

Evidence and optional classroom resources

Two-point linkage classroom reference

https://knowledge.carolina.com/labsheets/two-point-linkage-with-drosophila/

External resources remain on their publishers' websites. This packet contains original text and tables, not supplier worksheets. Chapter numbers are reading connections. No supplier or textbook publisher endorsement is implied.

Teacher pilot checklist

Confirm availability and preparation time; test controls; check units and observation windows; verify handling/disposal; choose class grouping and accessible alternatives; keep notes for revision.